The differential equations given below will arise frequently in various problems in this course. You should be able to solve all of them. In these equations, the b's are constants and f, g, P, and Q are arbitrary functions.

a. $y' = f(x)g(y)$

b. $y' + P(x)y = Q(x)$

c. $y'' + b^2y = 0$

d. $y'' - b^2y = 0$

e. $y'' = b_o + b_1x + b_2x^2$

f. $\frac{1}{x}\frac{d}{dx}(x\frac{dy}{dx}) = b_o + b_1x + b_2x^2$

g* $\frac{1}{x^2}\frac{d}{dx}(x^2\frac{dy}{dx}) + b^2y = 0$

h* $\frac{1}{x^2}\frac{d}{dx}(x^2\frac{dy}{dx}) - b^2y = 0$

i. $y'' + b_1y' + b_2y = 0$

j. $y'' + 2xy' = 0$

k. $\frac{d}{dx}[f(y)\frac{dy}{dx}] = 0$

l. $\frac{1}{x^2}\frac{d}{dx}(x^2\frac{dy}{dx}) = b_o + b_1x + b_2x^2$

* Try the substitution $y(x) = u(x)/x$

a) $y' = f(x)g(y) \rightarrow \frac{dy}{dx} = f(x)g(y) \rightarrow \frac{1}{g(y)}dy = f(x)dx \rightarrow \boxed{\ln g(y) = f'(x) + C}$

b) $y' + P(x)y = Q(x)$

$\mu = e^{\int P(x)dx} \rightarrow \mu(y' + P(x)y) = Q(x)\mu$

$y'(\mu y) = Q(x)\mu \rightarrow \mu y = \int Q(x)\mu\, dx + C$

$$\boxed{y = \frac{\int Q(x)\mu dx}{\mu} + C}$$

c) $y'' + b^2y = 0$ homogenous

$ar^2 + br + C = 0 \rightarrow r^2 + b^2 = 0 \rightarrow r^2 = -b^2 \rightarrow r = \sqrt{-b^2} \rightarrow r = \sqrt{b^2}\sqrt{-1} = bi$

$y = e^{\alpha x}(C_1\cos(\beta x) + C_2\sin(\beta x)) \rightarrow \boxed{y = C_1\sin(bx) + C_2\cos(bx)}$

d) $y'' - b^2y = 0$

$r^2 - b^2 = 0 \rightarrow r^2 = b^2 \rightarrow r = b$

$$\boxed{y = e^{bx}}$$

e) $y'' = b_o + b_1x + b_2x^2$

$y' = b_ox + \frac{1}{2}b_2x^2 + \frac{1}{3}b_2x^3 + c$

$$\boxed{y = \frac{1}{2}b_ox^2 + \frac{1}{6}b_1x^3 + \frac{1}{12}b_2x^4 + c}$$

f) $\frac{1}{x}\frac{d}{dx}(x\frac{dy}{dx}) = b_o + b_1x + b_2x^2$

$$\frac{d}{dx}\left(x\frac{dy}{dx}\right) = b_0 x + b_1 x^2 + b_2 x^3$$

$$x\frac{dy}{dx} = \frac{1}{2}b_0 x^2 + \frac{1}{3}b_1 x^3 + \frac{1}{4}b_2 x^4 + C$$

$$\frac{dy}{dx} = \frac{1}{2}b_0 x + \frac{1}{3}b_1 x^2 + \frac{1}{4}b_2 x^3 + \frac{C_1}{x}$$

$$\boxed{y = \frac{1}{4}b_0 x^2 + \frac{1}{9}b_1 x^3 + \frac{1}{16}b_2 x^4 + C_1 \ln x + C_2}$$

g) $\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{dy}{dx}\right) + b^2 y = 0 \qquad y = \frac{U(x)}{x}$

$$\frac{1}{x^2}\left(x^2 y'' + 2xy'\right) + b^2 y = 0 \rightarrow y'' + \frac{2}{x}y' + b^2 y = 0$$

w/ $y = \frac{U(x)}{x} \rightarrow y' = \frac{1}{x}U(x)' \rightarrow \frac{1}{x}U'(x) - \frac{1}{x^2}U(x)$

$$y'' = \frac{U''}{x} - \frac{2U'}{x^2} + \frac{2U}{x^3}$$

$$y'' + \frac{2}{x}y' = \frac{U''}{x} \Rightarrow$$

just like C

$$\frac{U''(x)}{x} + b^2\frac{U(x)}{x} = 0 \rightarrow U''(x) + b^2 U(x) = 0 \rightarrow r^2 + b^2 = 0$$

$r^2 = -b^2 \rightarrow r = bi$

$U(x) = C_1\cos(bx) + C_2\sin(bx) \leftarrow$ w/ $y = \frac{U(x)}{x} \rightarrow$

$$\boxed{y = \frac{C_1\cos(bx) + C_2\sin(bx)}{x}}$$

h) $\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{dy}{dx}\right) - b^2 y = 0$

$\frac{1}{x^2}\left(x^2y'' + 2xy'\right) - b^2 y = 0 \rightarrow y'' + \frac{2}{x}y' - b^2 y = 0$, $y = \frac{U(x)}{x} \rightarrow \frac{U'(x)}{x} - \frac{U'(x)}{x^2} \rightarrow \frac{U''(x)}{x} - \frac{2U'(x)}{x^2} + \frac{2U(x)}{x^3}$

plug in to get $\frac{U''(x)}{x} - b^2\frac{U(x)}{x} = 0 \rightarrow U''(x) - b^2 U(x) = 0 \rightarrow r^2 - b^2 = 0$

$r = b$

$U(x) = C_1 e^{bx} \rightarrow y = \frac{U(x)}{x} \rightarrow$

$$\boxed{y = C_1\frac{e^{bx}}{x}}$$

i) $y'' + b_1 y' + b^2 y = 0$ homogenous

$ar^2 + b_1 + C = 0 \rightarrow r^2 + br + b^2 = 0$

quadratic equation:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{-b \pm \sqrt{b^2 - 4(1)(b^2)}}{2} \rightarrow \frac{-b \pm \sqrt{b^2 - 4b^2}}{2}$$

$\alpha \quad \beta$

b

$-b \pm \sqrt{-2b^2} \quad -b \pm \sqrt{2b^2}\sqrt{-1} \quad -b \pm 2bi$

$$\frac{\quad}{2} \to \frac{\quad}{2} \to \frac{\quad}{2} \to -\frac{\quad}{2} \pm b i$$

$$\boxed{y = e^{-\frac{b}{2}x}\left(C_1 \cos(bx) + C_2 \sin(bx)\right)}$$

j) $y'' + 2xy' = 0$ ← reduction of order

$b = y' \to b' + 2xb = 0 \to b' = -2xb \to \frac{db}{dx} = -2xb$

$\frac{1}{b}db = -2x\,dx \to \ln|b| = -x^2 + C \to b = e^{-x^2 + C_1} \to y' = e^{-x^2+C}$

$y' = e^{-x^2+C_1} \to \boxed{y = \int e^{-x^2+C_1}dx + C_2}$

k) $\frac{d}{dx}\left[f(y)\frac{dy}{dx}\right] = 0$

$f(y)\frac{dy}{dx} = C_1 \to f(y)\,dy = C_1\,dx$

$$\boxed{\begin{array}{c} f'(y) = C_1 x + C_2 \\ y = \frac{1}{2}C_1 x^2 \end{array}}$$

l) $\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{dy}{dx}\right) = b_0 + b_1 x + b_2 x^2$

$\frac{d}{dx}\left(x^2\frac{dy}{dx}\right) = b_0 x^2 + b_1 x^3 + b_2 x^4$

$x^2\frac{dy}{dx} = \frac{1}{3}b_0 x^3 + \frac{1}{4}b_1 x^4 + \frac{1}{5}b_2 x^5 + C_1$

$\frac{dy}{dx} = \frac{1}{3}b_0 x + \frac{1}{4}b_1 x^2 + \frac{1}{5}b_2 x^3 + \frac{C_1}{x^2}$

$$\boxed{y = \frac{1}{6}b_0 x^2 + \frac{1}{12}b_1 x^3 + \frac{1}{20}b_2 x^4 - \frac{C_1}{2x^3} + C_2}$$